

Datasets Used

Dataset	Source	Role in system
mirpur12_ground_data.csv	Field survey / GPS verification	20 baseline points with coordinates, temperature, surface class, NDVI, traffic density, timestamps, and site images.
mirpur_unified_environmental_data.csv	Project-curated historical environmental observations	Longitudinal training and analysis dataset for Mirpur; cleaned to 11,627 rows after validation.
mirpur_calculated_environmental_indices.csv	Pipeline-generated derived data	Calculated Red/NIR proxies, NDVI, LST_C, and Temperature used by mapping, risk scoring, and

Dataset	Source	Role in system
		regression.
<code>mirpur_historical_temperatures.csv</code>	Historical temperature log	Annual trend input for the future temperature forecast endpoint.
<code>{region}_data_calculated.csv</code>	Simulated divisional presets	15-point demonstration layers for Dhaka, Sylhet, Rajshahi, and Chittagong generated with a fixed seed for reproducible scale-up tests.
<code>satellite_data/placeholders.js</code>	Regional geometry presets	Bounding vectors and region centers for the GIS layer and map centering logic.

Model Inventory

LinearRegression

Provider: scikit-learn. License: BSD-3-Clause. Used for the interpretable temperature-vs-NDVI baseline.

DecisionTreeRegressor

Provider: scikit-learn. License: BSD-3-Clause. Used to capture local nonlinear effects without pretrained weights.

RandomForestRegressor

Provider: scikit-learn. License: BSD-3-Clause. Used as the stronger ensemble comparator for site-level regression.

Hybrid residual forecasting stack

Provider: project code. License: repository license. Combines linear trend + residual tree models for long-horizon projections.

Pretrained Model Use

None. The AI component does not call any external pretrained or foundation model. All predictive models are trained locally from

project data and saved as metrics JSON artifacts.

Known Limitations

- Divisional regions are demonstration layers, not field-surveyed equivalents to Mirpur 12.

- Red, NIR, and LST values are proxy-estimated from surface/weather context when raw satellite bands are

unavailable.

- Metrics are fitted on project artifacts, so they should be read as decision-support signals rather than

operational forecasts.

- Geotagged photos and map popups should be shared responsibly; no personal data is intended, but

location disclosure can still matter.

Ethical Notes

- The system is optimized for urban planning support, not individual surveillance or enforcement.
- Proxy-derived thermal estimates should not be presented as ground instrument truth without calibration disclosure.
- Generalizing the Mirpur 12 pattern to the rest of Bangladesh requires caution because morphology, moisture, and coastal wind regimes differ substantially by region.

Prepared from the repo artifacts and the SciBlitz AI Challenge rulebook requirements. No pretrained foundation model is used in the AI
stack.